

SAMRAT MAJHI

AI/ML Engineer | Data Scientist

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Kolkata, India

SUMMARY

Final-year B.Tech CSE student with hands-on experience building production-grade AI systems — multilingual healthcare assistants, NLP classifiers, and generative AI chatbots. Proficient in Python, LangChain, LangGraph, FastAPI, Docker, and AWS/GCP deployment. Passionate about LLM applications, RAG architecture, and MLOps pipelines.

SKILLS

Languages & Libraries:
Python, JavaScript, SQL, Numpy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, Matplotlib, Seaborn.

Machine Learning & Data Science:
Data Cleaning, Model Training, Hyperparameter Tuning, Data Pipeline, Exploratory Data Analysis, A/B Testing.

GenAI & Deep Learning:
LangChain, LangGraph, RAG, Hugging Face, Vector Database, LLMs, RNN, CNN, LSTM, Neural Networks.

MLOps & Deployment:
Docker, FastAPI, AWS, GCP, CI/CD, MLflow, Git.

PROJECTS

ArogyaAI – Multilingual WhatsApp Multi-Agent Healthcare Assistant

|| [LiveApp](#) || [Code](#)

- Technologies Used:** Python, JavaScript,CSS, FastAPI, LangGraph, RAG, Twilio, Docker, AWS, EC2, Vector Database.
- Architected and deployed a **14-agent LangGraph** healthcare assistant on **AWS EC2**, handling **100+ concurrent conversations** across symptom triage, follow-up, disease guidance, and emergency escalation via **Twilio WhatsApp**.
 - Engineered a RAG pipeline over a curated vaccination knowledge base, **ensuring 100% of responses stayed grounded in verified medical source data** (zero fabricated facts), delivering context-aware multilingual guidance to users.
 - Designed an end-to-end **MLOps** architecture using **Docker, FastAPI, AWS EC2, and Vercel**, exposing scalable **REST APIs** and enabling continuous deployment of a production-ready AI healthcare platform.

User Intent Detection & Drift Monitoring System

|| [LiveApp](#) || [Code](#)

- Technologies Used:** Python, Scikit-learn, TF-IDF, LR, FastAPI, Render, Docker.
- Automated banking query routing with **94% intent classification accuracy**, cutting **manual triage effort by 30%**, by building a **TF-IDF + Logistic Regression** model across balance, loan, fraud, and card queries for downstream **NLP workflows**.
 - Reduced misrouted requests by 25%** by engineering a confidence-based drift detection framework that flags **low-confidence predictions (<0.75)** for early model degradation detection.
 - Built and deployed a production-ready **MLOps pipeline** with **FastAPI, Streamlit, Docker, Apache Airflow, and GitHub Actions CI/CD**, enabling automated retraining, **live monitoring**, and self-service **issue resolution**.

WORK HISTORY

- Quirky Connections Pvt Ltd : Generative AI Developer Intern || 03/2025 - 04/2025|| Remote
- Built and deployed **2 production-ready chatbots** with **PDF summarization and voice support** using **GPT-4/Gemini APIs**, achieving **sub-2-second** response time on **AWS EC2**.
 - Containerized with **Docker** and served via **FastAPI** REST backend on **AWS EC2** for reliable production deployment.
- Pinnacle Labs Pvt Ltd : Data Science Intern || 07/2025 - 08/2025 || Remote
- Trained **spaCy + ML classification models** for sentiment analysis and fake news detection, achieving **89% accuracy** on validation datasets across multiple model comparisons.
 - Built a **CNN-based image classification pipeline** using **TensorFlow**, with end-to-end data preprocessing, cleaning, and model evaluation — deployed on **GCP**.

EDUCATION

Bachelor of Technology in Computer Science Engineering (MAKAUT)	09/2023 – 07/2026
Pailan College of Management & Technology	CGPA – 8.6 Kolkata
Computer Science and Technology Diploma (WBSCTVESD)	08/2020 – 08/2023
Budge Budge Institute of Technology	CGPA - 7.4 Kolkata

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Data Science [🔗](#)
- AWS Certified Machine Learning - Specialty (MLS-C01) [🔗](#)

COMPETITIONS & HACKATHONS

- Amazon ML Challenge – 2025:** Developed a Smart Product Pricing model by processing 75,000+ product images and training deep learning models on Google Colab for large-scale data handling and optimization.
- Google Gen AI Hackathon - 2025:** Built a generative AI solution to support and improve youth mental wellness.